

Two Touchstones: the Discriminant and the Cubic Formula

A Road into Quadratic and Cubic Reciprocity

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Abstract

The discriminant of a quadratic and the formula for a cubic provide two familiar starting points for one arithmetic question: when does an equation have a solution after its coefficients are reduced modulo a prime? Quadratic residues lead to the Legendre symbol and quadratic reciprocity. Pure cubics $t^3 - a$ force the cube roots of unity into the discussion; their discriminant $-27a^2$ singles out $\mathbb{Q}(\sqrt{-3})$, whose ring of integers is the Eisenstein ring $\mathbb{Z}[\omega]$. There the cubic residue symbol and cubic reciprocity arise naturally. Hand calculations precede the code, and two colour-coded matrices make the contrasting symmetry laws visible.

1 The quadratic touchstone

For

$$ax^2 + bx + c = 0$$

the discriminant is

$$D = b^2 - 4ac.$$

Over $\mathbb{R}$, its sign distinguishes two real roots, a repeated root, and a complex-conjugate pair. Now let p be an odd prime with $p \nmid a$. Completing the square gives

$$(2ax + b)^2 \equiv D \pmod{p}.$$

The quadratic therefore has a root in $\mathbb{F}_p$ exactly when D is a square modulo p , including the possibility $D \equiv 0 \pmod{p}$.

Definition 1 (Legendre symbol). For an odd prime p ,

$$\left(\frac{u}{p}\right) = \begin{cases} 0, & p \mid u, \\ +1, & u \text{ is a nonzero square modulo } p, \\ -1, & u \text{ is not a square modulo } p. \end{cases}$$

Thus $\left(\frac{u}{p}\right)$ records whether $x^2 \equiv u \pmod{p}$ is solvable.

1.1 A first hand calculation

Take $p = 7$. Squaring the nonzero residues gives

x	1	2	3	4	5	6
$x^2 \bmod 7$	1	4	2	2	4	1

so the nonzero squares modulo 7 are 1, 2, 4. Hence

$$\left(\frac{2}{7}\right) = +1 \quad \text{because} \quad 3^2 \equiv 2 \pmod{7}, \quad \left(\frac{3}{7}\right) = -1.$$

Accordingly, $x^2 - 2 \equiv 0 \pmod{7}$ has the two solutions $x \equiv 3, 4$, while $x^2 - 3 \equiv 0 \pmod{7}$ has none.

1.2 Euler's criterion: the hand table becomes an algorithm

The table can be replaced by one exponentiation. For $p \nmid u$,

$$u^{(p-1)/2} \equiv \left(\frac{u}{p}\right) \pmod{p}. \quad (1)$$

To see why, choose a generator g of the cyclic group $\mathbb{F}_p^\times$ and write $u = g^k$. Then

$$u^{(p-1)/2} = g^{k(p-1)/2} \equiv (-1)^k \pmod{p}.$$

The element g^k is a square exactly when k is even. Equation (1) is Euler's criterion, the hinge between the residue table and the later code.

1.3 Quadratic reciprocity as an exchange law

For distinct odd primes p and q , quadratic reciprocity states

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}}. \quad (2)$$

The two symbols agree unless $p \equiv q \equiv 3 \pmod{4}$, when they have opposite signs.

For $p = 3$ and $q = 7$,

$$\left(\frac{3}{7}\right) = -1, \quad \left(\frac{7}{3}\right) = \left(\frac{1}{3}\right) = +1.$$

For $p = 3$ and $q = 5$,

$$\left(\frac{3}{5}\right) = -1, \quad \left(\frac{5}{3}\right) = \left(\frac{2}{3}\right) = -1.$$

The congruence classes modulo 4 determine whether the exchanged symbols agree; they do not determine either symbol separately.

1.4 A brief view of splitting

If d is a nonsquare integer, adjoining $\sqrt{d}$ to $\mathbb{Q}$ produces the quadratic field $\mathbb{Q}(\sqrt{d})$. For an odd prime p away from the field discriminant, the value $\left(\frac{d}{p}\right)$ records whether p splits into two prime factors or remains prime in the ring of integers of the field. The value $+1$ means split and -1 means inert. This is the number-field version of the elementary question whether $x^2 \equiv d \pmod{p}$ has a solution. We shall need only the special field $\mathbb{Q}(\sqrt{-3})$ later.

Listing 1: Euler's criterion computes the Legendre symbol.

```

1 def legendre(a: int, p: int) -> int:
2     """Return the Legendre symbol (a/p) for an odd prime p."""
3     a %= p
4     if a == 0:
5         return 0
6     value = pow(a, (p - 1) // 2, p)
7     return 1 if value == 1 else -1
8
9 residues_13 = sorted({k * k % 13 for k in range(1, 13)})
10 print(residues_13)           # [1, 3, 4, 9, 10, 12]
11 print(legendre(3, 13))      # +1, since 4^2 = 3 (mod 13)
12 print(legendre(3, 5))      # -1

```

2 The cubic touchstone

Cardano's formula begins with the general depressed cubic

$$t^3 + At + B = 0.$$

Its discriminant is

$$\Delta = -4A^3 - 27B^2,$$

and Cardano's expression contains

$$\sqrt{\left(\frac{B}{2}\right)^2 + \left(\frac{A}{3}\right)^3} = \sqrt{-\frac{\Delta}{108}}.$$

Over $\mathbb{R}$, the sign of the quantity under this square root distinguishes one real root, a repeated root, and the casus irreducibilis of three distinct real roots reached through complex intermediate quantities.

Boundary marker 1 (Why we now restrict the question). For a general cubic, knowing whether Δ is a square modulo p gives useful information about its factorisation, although it does not by itself distinguish complete splitting from irreducibility. Completing that general story leads beyond elementary cubic residues and towards Galois and class field theory. From this point onward we study the pure cubic $t^3 - a$. Here the cubic residue symbol answers exactly the question we ask.

2.1 The pure cubic forces the field $\mathbb{Q}(\sqrt{-3})$

For

$$f(t) = t^3 - a$$

the discriminant is

$$\Delta = -27a^2.$$

When $a \neq 0$,

$$\mathbb{Q}(\sqrt{\Delta}) = \mathbb{Q}(\sqrt{-3}).$$

The same field appears directly from the roots. If $r^3 = a$ and

$$\omega = e^{2\pi i/3}, \quad \omega^2 + \omega + 1 = 0,$$

then

$$t^3 - a = (t - r)(t - \omega r)(t - \omega^2 r).$$

One cube root brings the others with it only after ω has been adjoined. The ring of integers of $\mathbb{Q}(\sqrt{-3})$ is

$$\mathbb{Z}[\omega] = \{x + y\omega : x, y \in \mathbb{Z}\},$$

the ring of Eisenstein integers. Its norm is

$$N(x + y\omega) = (x + y\omega)(x + y\omega^2) = x^2 - xy + y^2.$$

Thus the discriminant and the roots of the pure cubic independently lead to the same arithmetic setting.

2.2 Cubes modulo an ordinary prime

Modulo 7,

x	1	2	3	4	5	6
$x^3 \bmod 7$	1	1	6	1	6	6

so the nonzero cubes are 1 and 6. The map $x \mapsto x^3$ on $\mathbb{F}_p^\times$ has a kernel of size 3 when $p \equiv 1 \pmod{3}$. Consequently its image contains $(p-1)/3$ elements, and the nonzero elements fall into three cubic classes.

If instead $p \equiv 2 \pmod{3}$, then 3 is coprime to $p-1$. Cubing is a bijection on $\mathbb{F}_p^\times$, so every nonzero element is a cube. The interesting three-way distinction therefore occurs at primes $p \equiv 1 \pmod{3}$.

3 The cubic residue symbol in $\mathbb{Z}[\omega]$

A rational prime $p \equiv 1 \pmod{3}$ splits in the Eisenstein integers:

$$p = \pi \bar{\pi}, \quad N\pi = p.$$

There are two conjugate primes above p . This choice will matter to the label ω or ω^2 , so we shall make it explicit.

Definition 2 (Primary and canonical primary primes). Let $\pi = a + b\omega$ with $3 \nmid N\pi$. It is primary when

$$\pi \equiv 2 \pmod{3}, \quad \text{equivalently} \quad a \equiv 2 \pmod{3}, \quad b \equiv 0 \pmod{3}.$$

Among the six associates $\pi, -\pi, \omega\pi, -\omega\pi, \omega^2\pi, -\omega^2\pi$, exactly one is primary. For each rational prime $p \equiv 1 \pmod{3}$, however, both a prime π above p and its conjugate $\bar{\pi}$ are primary. In this paper the canonical primary prime above p means the one written $a + b\omega$ with $b > 0$.

For example,

$$7 = (2 + 3\omega)(2 + 3\omega^2).$$

In the $a + b\omega$ notation the two primary primes are

$$2 + 3\omega \quad \text{and} \quad -1 - 3\omega.$$

Our $b > 0$ convention selects $2 + 3\omega$.

Definition 3 (Cubic residue symbol). Let π be an Eisenstein prime with $N\pi \neq 3$, and suppose $\pi \nmid \alpha$. The cubic residue symbol is the unique element of $\{1, \omega, \omega^2\}$ satisfying

$$\alpha^{(N\pi-1)/3} \equiv \left(\frac{\alpha}{\pi}\right)_3 \pmod{\pi}.$$

The symbol equals 1 precisely when α is a cube modulo π .

This is the cubic analogue of Euler's criterion. The values ω and ω^2 distinguish the two non-cube classes rather than merely reporting "not a cube."

3.1 A cubic symbol by hand

Let $\pi = 2 + 3\omega$, the canonical primary prime above 7. In the quotient $\mathbb{Z}[\omega]/(\pi) \cong \mathbb{F}_7$,

$$2 + 3\omega \equiv 0 \implies \omega \equiv -2 \cdot 3^{-1} \equiv 4 \pmod{7}.$$

Thus $\omega^2 \equiv 2 \pmod{7}$. For $\alpha = 2$,

$$2^{(N\pi-1)/3} = 2^2 \equiv 4 \equiv \omega \pmod{\pi},$$

so

$$\left(\frac{2}{2+3\omega} \right)_3 = \omega.$$

Likewise $3^2 \equiv 2 \equiv \omega^2 \pmod{7}$, giving

$$\left(\frac{3}{2+3\omega} \right)_3 = \omega^2.$$

3.2 Cubic reciprocity in numbers

For primary Eisenstein primes π and λ with distinct norms, cubic reciprocity states

$$\left(\frac{\pi}{\lambda} \right)_3 = \left(\frac{\lambda}{\pi} \right)_3. \quad (3)$$

Take the canonical choices

$$\pi = 2 + 3\omega, \quad N\pi = 7, \quad \lambda = -1 + 3\omega, \quad N\lambda = 13.$$

Modulo π , $\omega \equiv 4 \pmod{7}$, so

$$\lambda \equiv -1 + 3(4) \equiv 4 \pmod{7}, \quad \lambda^2 \equiv 2 \equiv \omega^2 \pmod{\pi}.$$

Hence $\left(\frac{\lambda}{\pi} \right)_3 = \omega^2$.

Modulo λ , $\omega \equiv 3^{-1} \equiv 9 \pmod{13}$. Therefore

$$\pi \equiv 2 + 3(9) \equiv 3 \pmod{13}, \quad \pi^4 \equiv 3 \equiv \omega^2 \pmod{\lambda}.$$

Thus

$$\left(\frac{\lambda}{\pi} \right)_3 = \left(\frac{\pi}{\lambda} \right)_3 = \omega^2.$$

This equality is the contrast that the matrices below are built to make visible.

4 Computing without hiding the conventions

If $\pi = a + b\omega$ has norm p , then in $\mathbb{Z}[\omega]/(\pi) \cong \mathbb{F}_p$,

$$\omega \equiv -ab^{-1} \pmod{p} =: r_\pi.$$

The reduction map is therefore

$$\varphi_\pi(x + y\omega) = x + yr_\pi \pmod{p}.$$

The code below returns the exponent 0, 1, or 2 in ω^k , and names the function accordingly. It also uses an explicit extended Euclidean algorithm, so it does not depend on the modular-inverse syntax introduced in Python 3.8.

Listing 2: Canonical primary primes and cubic-symbol exponents.

```

1 from math import isqrt
2
3 def inverse_mod(a: int, m: int) -> int:
4     """Return  $a^{-1}$  modulo m using the extended Euclidean algorithm."""
5     old_r, r = a % m, m
6     old_s, s = 1, 0
7     while r:
8         q = old_r // r
9         old_r, r = r, old_r - q * r
10        old_s, s = s, old_s - q * s
11    if old_r != 1:
12        raise ValueError("inverse does not exist")
13    return old_s % m
14
15 def associates(a: int, b: int):
16     """The six unit multiples of  $a+b\omega$ , as coefficient pairs."""
17     return [(a, b), (-b, a - b), (b - a, -a),
18             (-a, -b), (b, b - a), (a - b, a)]
19
20 def primary_associate(a: int, b: int):
21     """Return the unique primary element among the six associates."""
22     for x, y in associates(a, b):
23         if x % 3 == 2 and y % 3 == 0:
24             return x, y
25     raise ValueError("no primary associate")
26
27 def norm_representation(p: int):
28     """Find one  $a, b$  with  $a^2 - a*b + b^2 = p$  using a one-dimensional search."""
29     for b in range(1, isqrt(4 * p // 3) + 2):
30         disc = 4 * p - 3 * b * b
31         if disc < 0:
32             break
33         root = isqrt(disc)
34         if root * root == disc and (b + root) % 2 == 0:
35             return (b + root) // 2, b
36     raise ValueError("p has no Eisenstein norm representation")
37
38 def conjugate(z):
39     """Conjugate  $a+b\omega$ :  $a+b\omega^2 = (a-b) - b\omega$ ."""
40     a, b = z
41     return a - b, -b
42
43 def canonical_primary(p: int):
44     """Choose the primary prime  $a+b\omega$  above p with  $b > 0$ ."""
45     a, b = primary_associate(*norm_representation(p))
46     if b < 0:
47         a, b = conjugate((a, b))
48     return a, b
49
50 def cubic_symbol_exponent(alpha, pi, p: int) -> int:
51     """Return  $k$  in  $\{0, 1, 2\}$  for  $(\alpha/\pi)_3 = \omega^k$ ."""
52     a, b = pi
53     r = (-a * inverse_mod(b, p)) % p # image of  $\omega$  modulo  $\pi$ 
54     x, y = alpha
55     reduced = (x + y * r) % p
56     if reduced == 0:
57         raise ValueError("symbol is zero because pi divides alpha")

```

```

58 value = pow(reduced, (p - 1) // 3, p)
59 for k, target in enumerate((1, r, r * r % p)):
60     if value == target:
61         return k
62     raise ValueError("unexpected cubic-character value")

```

5 Two reciprocity matrices

Both matrices index rows and columns by primes. The diagonal is grey because the corresponding symbol is zero rather than one of the nonzero character values. Figure 1 uses a colour-blind-safe blue/orange pair. Its asymmetry records the correction term in quadratic reciprocity. Figure 2 uses three distinguishable colours and is symmetric.

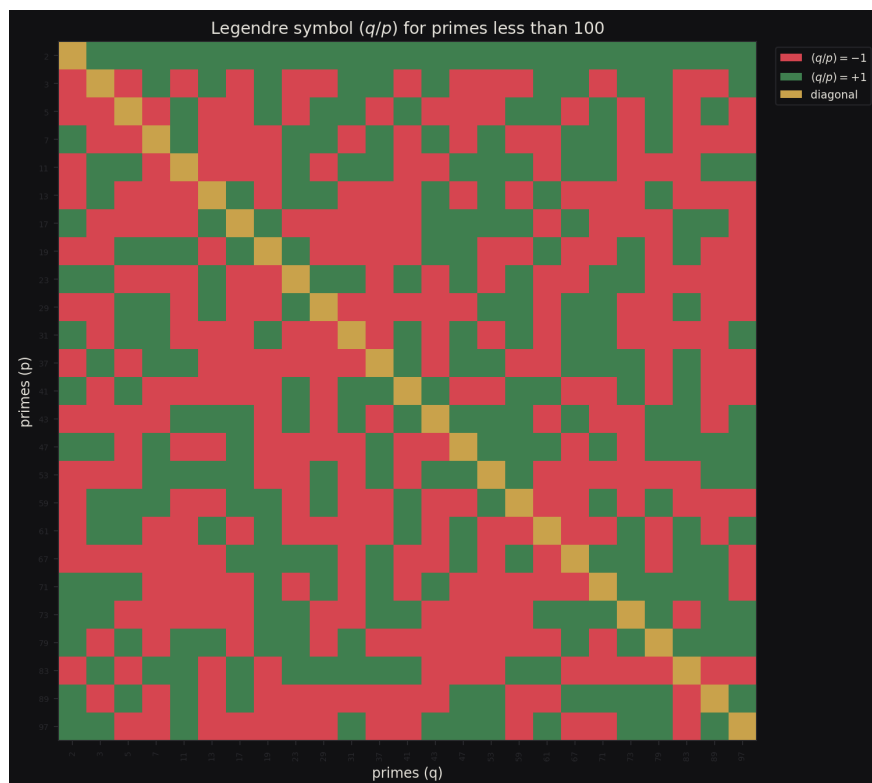


Figure 1: Legendre symbols (q/p) for distinct odd primes below 100. The value in a cell depends on the primes themselves, not merely on their congruence classes modulo 4. Those congruence classes determine the disagreement pattern between the transposed cells (q/p) and (p/q) . Blue and orange represent +1 and -1; grey marks the diagonal.

Listing 3: Checking reciprocity and the conjugate convention.

```

1 def primes_below(n: int):
2     """A small sieve, sufficient for constructing the displayed matrix."""
3     sieve = [True] * n
4     sieve[0:2] = [False, False]
5     for k in range(2, isqrt(n - 1) + 1):
6         if sieve[k]:
7             sieve[k * k:n:k] = [False] * (((n - 1 - k * k) // k) + 1)
8     return [k for k in range(n) if sieve[k]]
9
10 primes = [p for p in primes_below(100) if p % 3 == 1]

```

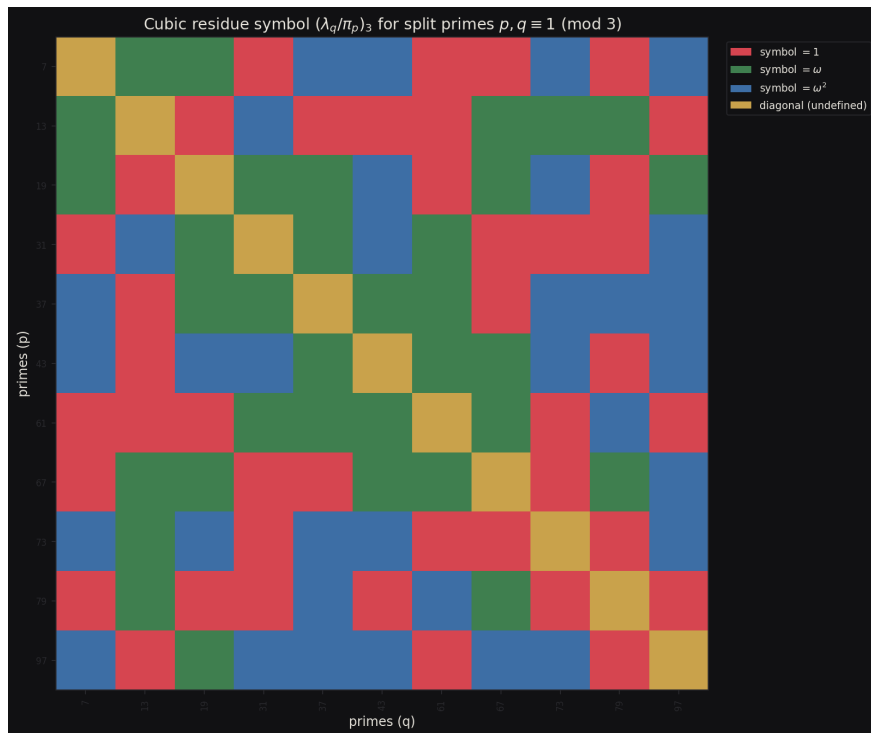


Figure 2: Cubic residue symbols $(\lambda_q/\pi_p)_3$ for primes $p, q \equiv 1 \pmod{3}$ below 100. For every rational prime, π_p is the canonical primary prime $a + b\omega$ with $b > 0$. Choosing all conjugate primes instead conjugates the labels $\omega \leftrightarrow \omega^2$ while preserving the symmetry. Grey marks the diagonal.

```

11 chosen = {p: canonical_primary(p) for p in primes}
12
13 def matrix_is_symmetric(data):
14     return all(
15         cubic_symbol_exponent(data[q], data[p], p)
16         == cubic_symbol_exponent(data[p], data[q], q)
17         for p in primes for q in primes if p != q
18     )
19
20 conjugates = {p: conjugate(chosen[p]) for p in primes}
21 print(matrix_is_symmetric(chosen))      # True
22 print(matrix_is_symmetric(conjugates))  # True
23
24 # Simultaneous conjugation swaps exponents 1 and 2 and fixes 0.
25 labels_conjugate = all(
26     cubic_symbol_exponent(conjugates[q], conjugates[p], p)
27     == (-cubic_symbol_exponent(chosen[q], chosen[p], p)) % 3
28     for p in primes for q in primes if p != q
29 )
30 print(labels_conjugate)                  # True

```

5.1 Why one law has a twist and the other does not

Quadratic reciprocity contains the explicit correction factor in (2). It forces disagreement precisely when both primes are 3 modulo 4. Cubic reciprocity for primary Eisenstein primes has the cleaner form (3).

The primary convention is essential, although the absence of a correction term should not be explained merely by counting units. The Eisenstein field already contains the cube roots of unity, and the full explanation belongs to the local and global arithmetic of that field. In more advanced language it is expressed

through cubic Hilbert symbols and their product formula at the ramified places. Developing that theory would take us well beyond the present route. The honest conclusion here is narrower: after primary normalisation, Eisenstein's cubic reciprocity law is symmetric, while Gauss's quadratic law retains its sign.

6 Exercises

1. List the nonzero squares modulo 11 and use your list to evaluate $\left(\frac{2}{11}\right)$, $\left(\frac{3}{11}\right)$ and $\left(\frac{5}{11}\right)$.
2. Use Euler's criterion to calculate $\left(\frac{3}{13}\right)$. Then exhibit a square root of 3 modulo 13.
3. Without listing all squares, use quadratic reciprocity to compare $\left(\frac{11}{19}\right)$ and $\left(\frac{19}{11}\right)$.
4. Explain why the homomorphism $x \mapsto x^3$ on $\mathbb{F}_p^\times$ has exactly $(p-1)/3$ values when $p \equiv 1 \pmod{3}$, and why it is a bijection when $p \equiv 2 \pmod{3}$.
5. Verify that $13 = N(-1 + 3\omega)$, show that $-1 + 3\omega$ is primary, and identify its conjugate primary prime.
6. Working modulo $\pi = 2 + 3\omega$, evaluate $\left(\frac{4}{\pi}\right)_3$ and $\left(\frac{5}{\pi}\right)_3$ by hand.
7. Replace $\pi = 2 + 3\omega$ by its conjugate $-1 - 3\omega$ in the preceding exercise. Explain why the answers are complex conjugates of the original symbol values.

7 Closing perspective

The appearance of -3 is now structural rather than decorative. For a pure cubic $t^3 - a$, the discriminant is $-27a^2$, so its quadratic subfield is $\mathbb{Q}(\sqrt{-3})$. Factoring the same polynomial requires the three cube roots $r, \omega r, \omega^2 r$, which again introduces $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$. Its ring of integers $\mathbb{Z}[\omega]$ is therefore the natural home of cubic residuosity. The quadratic discriminant and the cubic roots have pointed to the same field from two directions.

References

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